COS40007 Artificial Intelligence for Engineering

Week 3 Studio Activities

ILO	Understand Advanced Data Preprocessing, Feature Engineering & Model Development
Aim	• Understand how to develop a Machine Learning model and compare different features. • Learn how to perform feature selection and dimensionality reduction. • Learn how to improve the performance of the ML model through hyperparameter tuning.
Resources	Books: 1. Prosise, Jeff. Applied machine learning and AI for engineers. "O'Reilly Media, Inc," 2022. 2. Raschka, Sebastian, Yuxi Hayden Liu, and Vahid Mirjalili. Machine Learning with PyTorch and Scikit-Learn: Develop machine learning and deep learning models with Python. Packt Publishing Ltd, 2022. Web Resources: 1. https://www.kaggle.com/code/faressayah/support-vector-machine-pca-tutorial-for-beginner 2. https://www.geeksforgeeks.org/svm-hyperparameter-tuning-using-gridsearchcv-ml/

Prerequisite: Completed Studio Activities on EDA (Weeks 1 & 2)
Dataset: Use the dataset from last week.
Prerequisites: Completed Studio 2 (Data Preprocessing)
Learning Objectives:

- Develop and compare models using different feature sets
- Understand the impact of preprocessing on model performance
- Apply hyperparameter tuning
- Use feature selection and dimensionality reduction
- Compare results across multiple modelling approaches

➢ **You will not use the Studoi 2 example dataset for any portfolio assessment(s) or group task(s).**

➢ **You need to apply these steps to your datasets for submitting Portfolio Assessment 2.**

In Studio 2, you prepared multiple versions of your dataset:

- Raw features
- Normalised features
- Features with engineering
- Selected features

Now, you will use these datasets to train models and understand how each preprocessing step affects performance. **This studio works for BOTH regression and classification tasks - choose the appropriate model type for your problem.**

Quick Reference: Regression vs Classification

If Your Problem Is...	Use This Instead
Classification (predicting categories)	Keep all instructions as written
Regression (predicting numbers)	Replace "Decision Tree Classifier" → "Decision Tree Regressor"
	Replace "SVM" (SVC) → "SVR" (Support Vector Regressor)
	Replace "accuracy" → "R^2 score" or "RMSE"
	Replace classification metrics with regression metrics.

Studio Activity 1: Model Development and Comparison with Decision Tree

Develop Decision Tree models using five different feature sets from Studio 2 and compare their performance.
Note that all cases have a 70-30% split for training and validation data, and compute the accuracy of each model.

1. converted_concrete.csv (all features without normalisation and without composite features)
2. normalised_concrete.csv (all features with normalisation and without composite features)
3. features_concrete.csv (all features with normalisation and containing composite features)
4. selected_feature_concrete.csv (selected features with normalisation)
5. selected_converted_concrete.csv (selected feature without normalisation) [here select only 'Cement', 'Water', 'Superplastic' and 'Age' as a feature]

Studio Activity 2: First Summary Table with Findings

You will obtain five different accuracy values once you develop the decision tree classifier using five different feature sets.
Summarise our outcome in terms of accuracy using the following table.

Model 1 [accuracy in %]	Model 2 [accuracy in %]	Model 3 [accuracy in %]	Model 4 [accuracy in %]	Model 5 [accuracy in %]

Answer these questions:

1. Which feature set performed best? Why do you think that is?
2. Did normalisation improve performance? Compare Model 1 vs Model 2.
3. Did feature engineering help? Compare Model 2 vs Model 3.
4. Was feature selection effective? Compare Model 3 vs Model 4.
5. Is normalisation always necessary? Compare Model 4 vs Model 5.
6. What pattern do you observe across the five models?

Studio Activity 3: Improve your model with SVM (or SVR)

1. First, shuffle the data and save it in another CSV file.
2. Train an SVM model using a 70%:30% split of the training and test sets, and measure the model's accuracy.
3. Next, perform both 5-fold cross-validation and measure accuracy.

Sample code given.

Save the classification accuracy of the above 2 cases.

Studio Activity 4: Hyperparameter Tuning

Now, use this reference link to perform hyperparameter tuning on your dataset.

1. Try different kernels and select one, such as `an SVF or an RBF`.
2. Use `GridSearchCV` to identify optimal hyperparameter values.
3. Now use the optimal values identified in `GridSearchCV` to update your SVM model in Activity 3 and obtain classification accuracy for both train-test split and 5-fold cross-validation.

Studio Activity 5: Feature Selection

Use *K*-Best feature selection to identify the most important features and retrain the model (described here), and generate a result of another 2 SVM models with

1. 70/30 train/test set split with hyperparameter tuning (using values obtained in Activity 4)
2. 5-fold class validation with hyperparameter tuning (using values obtained in Activity 4)

Studio Activity 6: Dimensionality Reduction with PCA only

Use Principal Component Analysis (PCA) to reduce the dimension of your data. Take the first 10 principal components as features and train 2 SVM models again.

1. 70/30 train/test set split with hyperparameter tuning (using values obtained in activity 4)
2. 5-fold class validation with hyperparameter tuning (using values obtained in activity 4)

Sample steps:
1. `pca = PCA().fit(X)` [available in from sklearn.decomposition import PCA]
2. Now, to take 10 (or choose based on the dataset) principal components, use *pca—components list.*
3. Then, the 10 (or choose based on the dataset) principal components will be used as the new X for training the SVM model.

Studio Activity 7: Final Summarisation

Prepare a summary table showing the accuracy values for the SVM models you developed. Also, visualise the final comparison. Answer the questions below.

1. Which preprocessing technique (normalisation, feature engineering, feature selection, or PCA) had the biggest impact on your model performance?
2. Did hyperparameter tuning significantly improve your SVM model? By how much?
3. Compare Decision Tree (Activity 1) vs SVM (Activity 7) - which performed better for your dataset?
4. Was there a trade-off between model performance and interpretability?

SVM model	Train-test split	Cross-validation
Original features	XX%	XX%
With hyper-parameter tuning	XX%	XX%
With feature selection and hype parameter tuning	XX%	XX%
With PCA and hyperparameter tuning	XX%	XX%